TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 22

5-MINUTE REVISION

Analysis of Covariance

Combining ANOVA with regression: covariates, the common-slope model, adjusted means and the adjusted F test.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

KAU Me-LON · www.agrinoteskau.com · Dept. of Agricultural Statistics, College of Agriculture, Vellayani

THE IDEA TO KEEP

Adjust for a relevant covariate measured before treatment.

REMEMBER

ANCOVA combines regression and ANOVA. Check linearity, comparable covariate slopes and the role of the covariate in the design.

APPLY IT

Initial plant height may improve precision when comparing final biomass; a measurement caused by the treatment needs different causal reasoning.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 What is ANCOVA?

ANCOVA combines **analysis of variance** with **regression adjustment** on a **covariate**: a measured explanatory variable.

Agri examples: initial plant stand, initial weight of animals, pre-treatment pest count, previous-season yield.

A covariate must be **related to the response** and **not caused by the treatment**. Ideally measured **before** treatment (pre-treatment covariate).

2 Blocking vs ANCOVA

	Blocking	ANCOVA
When	Before the experiment (layout)	During analysis
Handles	Predictable variation (soil, gradient)	Variation in a measured covariate
Aim	Both reduce experimental error	

Precision may improve but is **not guaranteed**.

3 Model & key quantities (CRD)

Common-slope model

$$Y_{ij} = \mu + \tau_i + \beta(X_{ij} - \bar{X}) + e_{ij}$$

Slope

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

Adjusted error SS

$$E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

Adjusted MSE

$$MSE_{adj} = \frac{E}{n - v - 1}$$

E_{xx}, E_{xy}, E_{yy} = within-treatment (error) sums of squares and products. RBD error df: $(r-1)(v-1)-1$.

4 Degrees of freedom

Fitting one covariate slope **uses one error df**.

CRD	$n - v - 1$
RBD	$(r-1)(v-1) - 1$

5 Adjusted means & F test

Adjusted treatment mean

$$\bar{Y}_{i,adj} = \bar{Y}_i - \hat{\beta}(\bar{X}_i - \bar{X})$$

Adjusted F

$$F = \frac{(E_1 - E)/(v-1)}{MSE_{adj}}$$

E_1: residual SS of the reduced model (no treatments, covariate kept); E: full model. Means are compared at a **common covariate value**.

6 Assumptions

- ✓ The usual ANOVA assumptions: independent errors, common variance, normality for exact tests
- ✓ **Linear** relationship between response and covariate
- ✓ **Equal slopes** across treatments (check treatment × covariate interaction)
- ✓ Covariate not affected by the treatments
- ✓ Error mean is zero

7 Steps of the analysis

1. Compute X, Y and XY SS for
2. Get E_{xx}, E_{xy}, E error
3. Estimate $\hat{\beta} = E_{xy}/E_{xx}$
4. Adjusted MSE_a error SS and
5. Adjusted treatment SS: $E_1 - E$
6. Adjusted F test
7. Compute adjusted means
8. Compare with pair- specific SEs

! Common mistakes

- ✗ Using a covariate the treatment affects.
- ✗ Forgetting the extra df lost.
- ✗ Assuming adjustment turns observational data into an experiment.

? Fill in the blanks

1. Initial plant stand counted before applying treatments is a suitable ——. **[covariate]**
2. With E_xy = 30 and E_xx = 60, the slope b = ——. **[0.5]**
3. Treatment mean 50, its covariate mean 12, overall covariate mean 10 and b = 0.5 give an adjusted mean of ——. **[49]**
4. A CRD with 20 observations, 4 treatments and one covariate has —— error df. **[15]**
5. A covariate must not be —— by the treatments. **[affected]**
6. The common-slope model assumes the treatment regression lines are ——. **[parallel]**